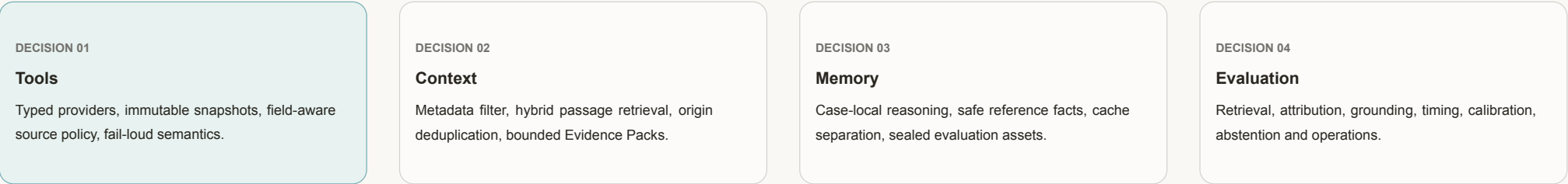
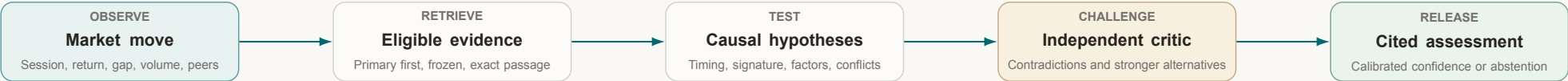


# ASX Unusual Trading Investigation Agent

A point-in-time, evidence-grounded system for explaining unusual moves in ASX-listed equities.

Deterministic market truth constrains model reasoning. Evidence and calibration determine what may be claimed.

Input: an ASX ticker and an ASX date. Output: a confidence-rated explanation in AUD and AEST/AEDT, with every material claim linked to evidence and every limitation visible.

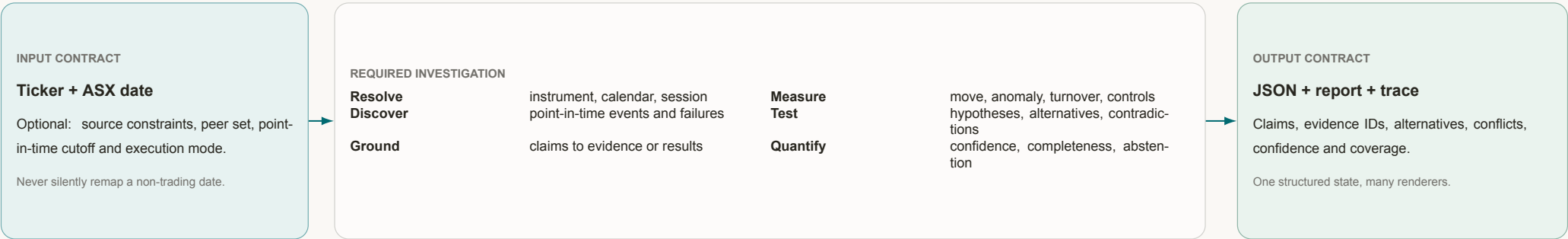


**STATUS BOUNDARY** This dossier specifies the target architecture. The repository now contains a runnable MVP with API, recorded and live gateways, Gemini synthesis, 12 passing Python tests and one passing recorded regression case. Confidence remains uncalibrated; no blind performance result exists.

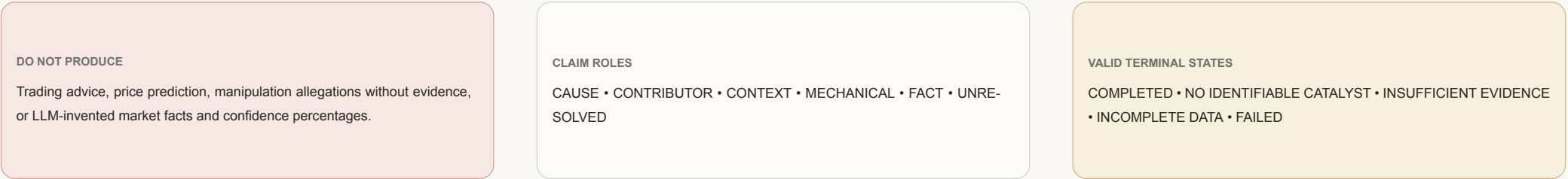
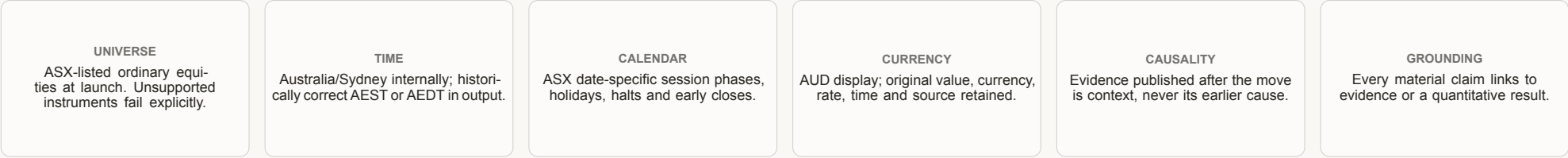
Basis: repository design specification, master development plan, frontend design baseline and verified repository inspection [R9-R12].

# What the product must prove

The answer must be inspectable, reproducible and allowed to end in uncertainty.



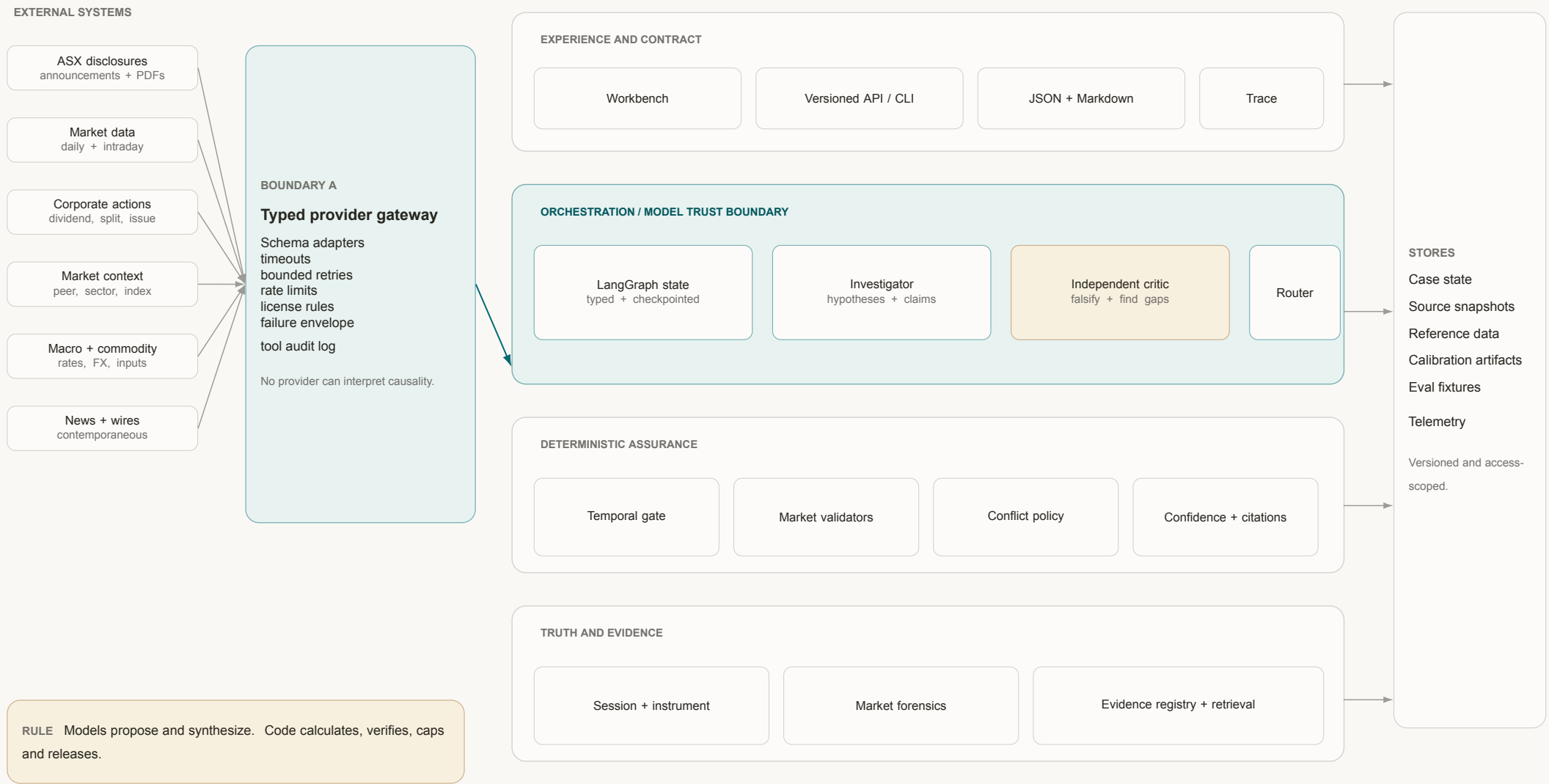
## Non-negotiable constraints



ASX session phases use Sydney time; normal continuous trading ends at 16:00 before the closing auction [R1-R2]. Continuous disclosures are primary issuer evidence [R3].

# Small agent surface, strong assurance core

Two reasoning nodes operate inside deterministic market, evidence and release boundaries.

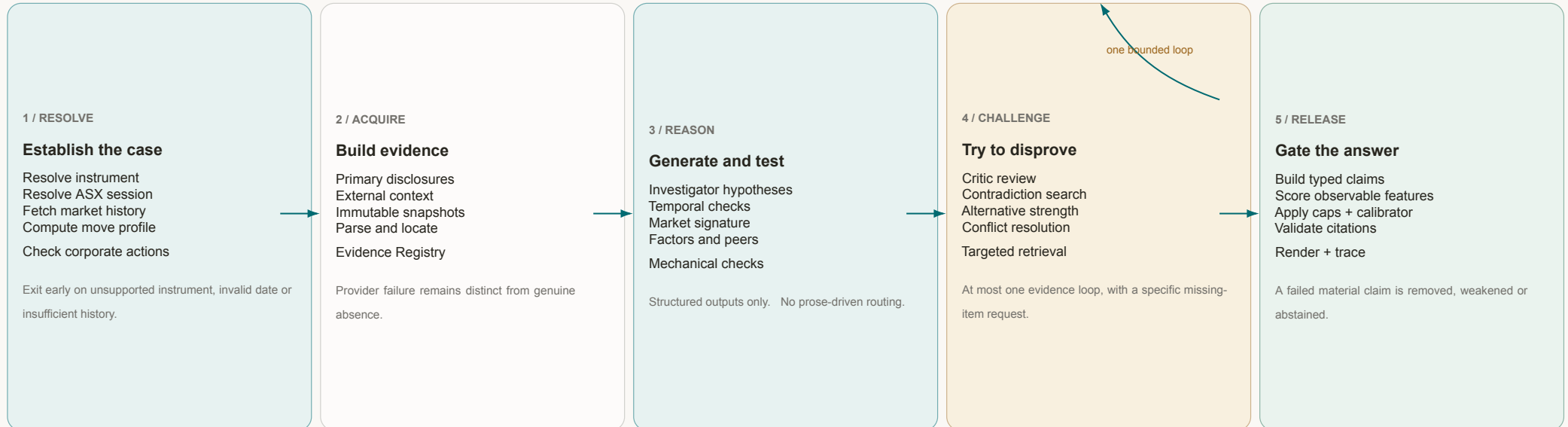


Architecture baseline [R9-R10]. LangGraph checkpoints support versioned graph-state persistence and resumption [R8].

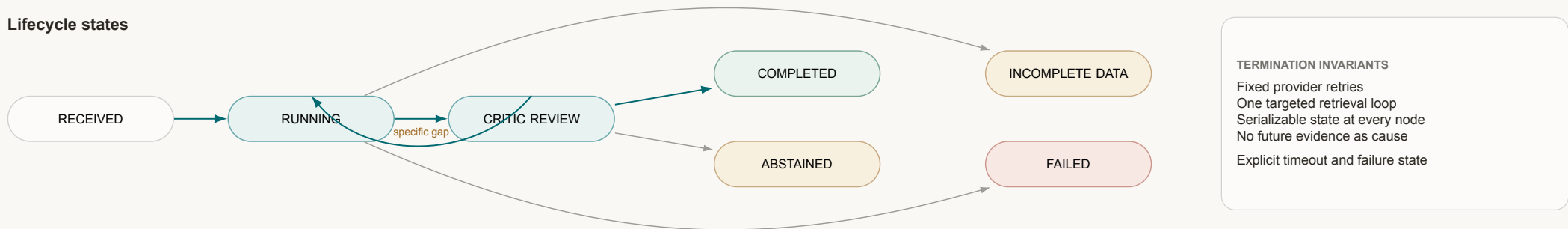
# Investigation graph and state machine

A bounded workflow separates discovery, validation, criticism and release.

## UML activity flow



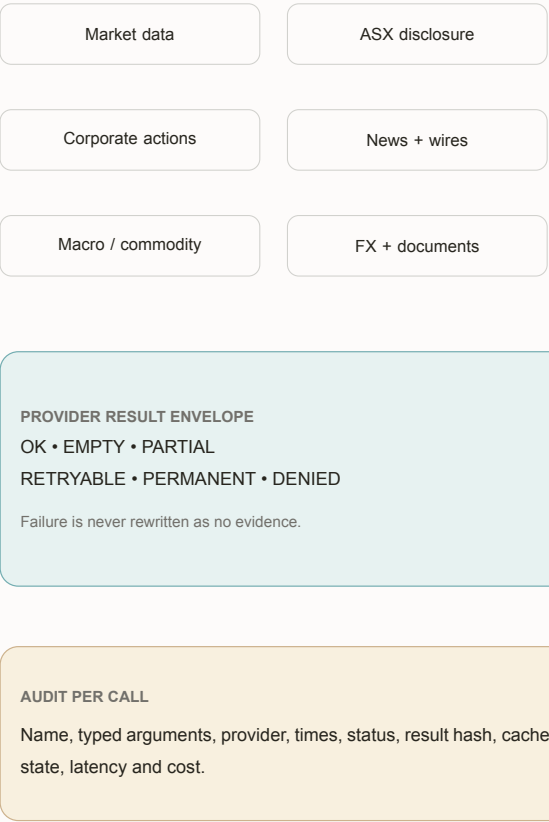
## Lifecycle states



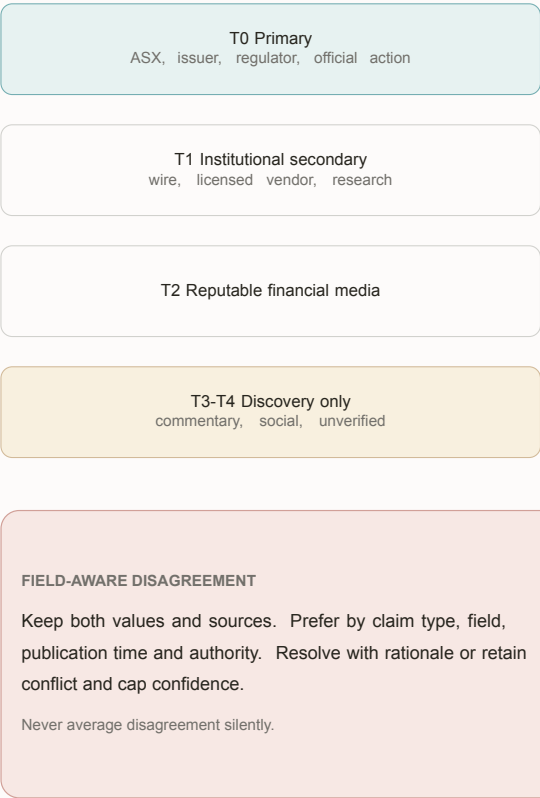
# From provider call to bounded Evidence Pack

Data enters through typed contracts; documents remain untrusted until normalized, filtered and registered.

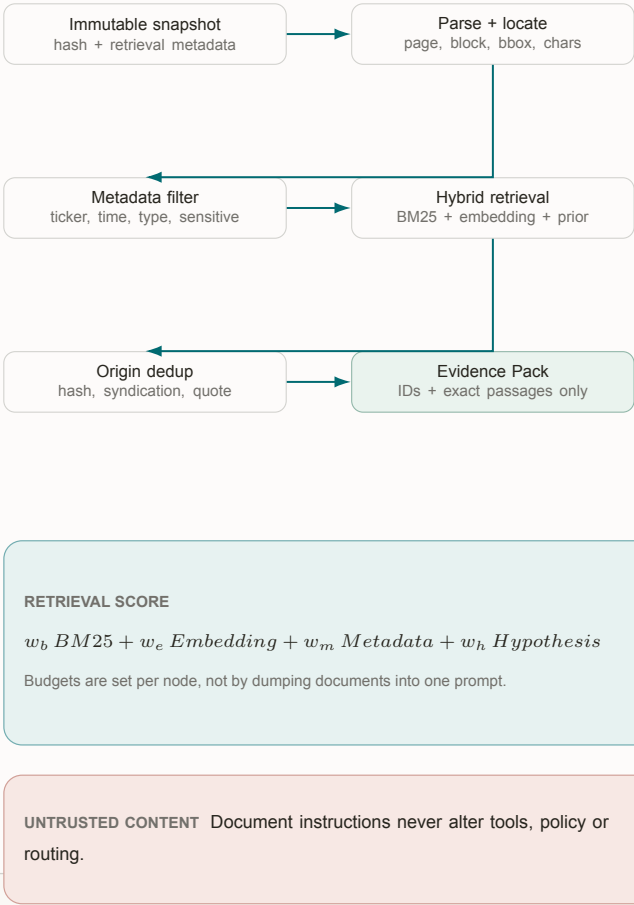
### A. Typed tools



### B. Source and conflict policy



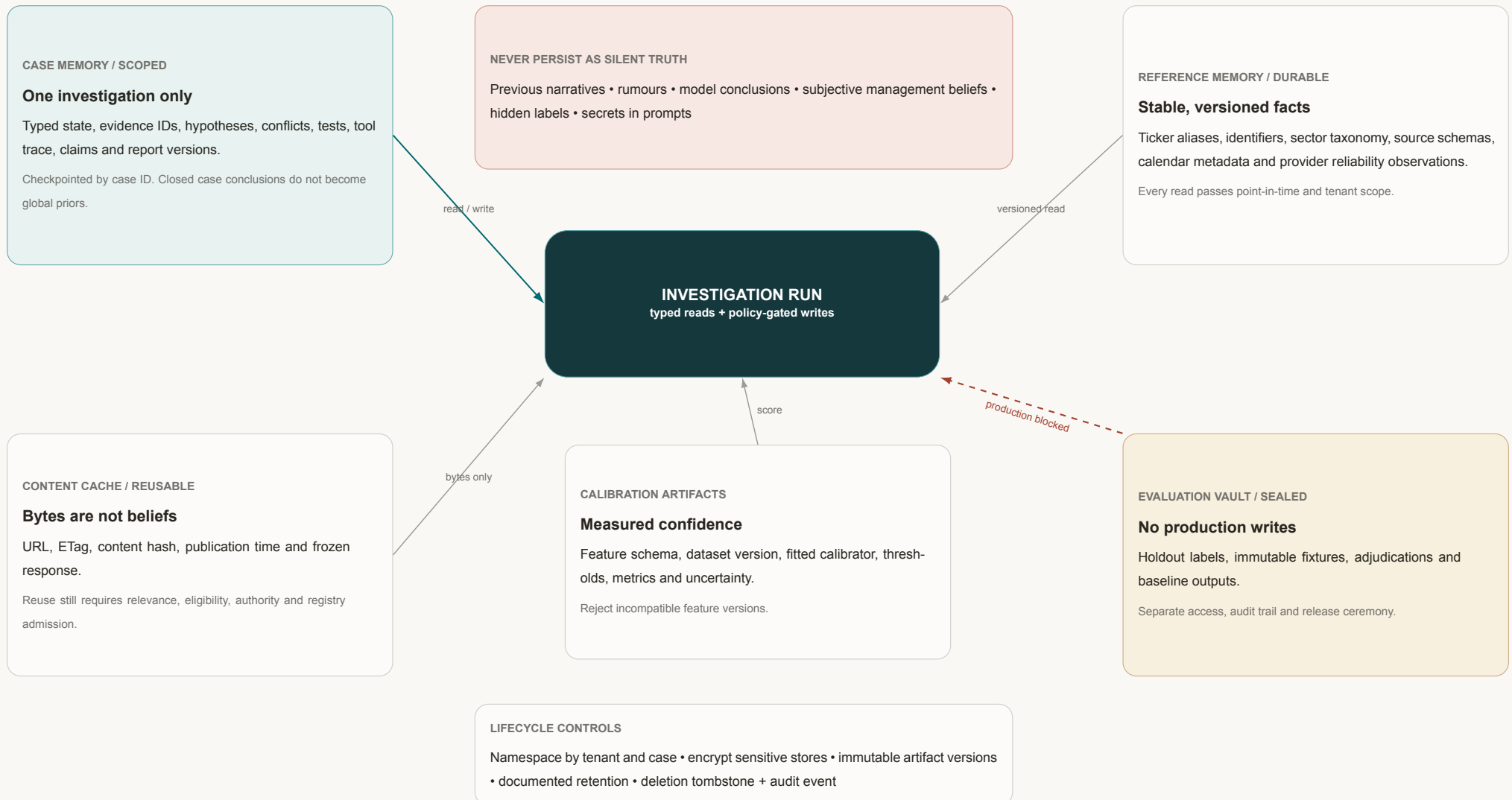
### C. Context pipeline



ASX disclosures and session sources [R1-R3]. Indirect prompt injection can arrive through retrieved content, tool output and memory [R6]. Architecture baseline [R9-R10].

# Persist facts and artifacts, not conclusions

Memory is split by scope, authority and lifecycle so one case cannot silently bias the next.

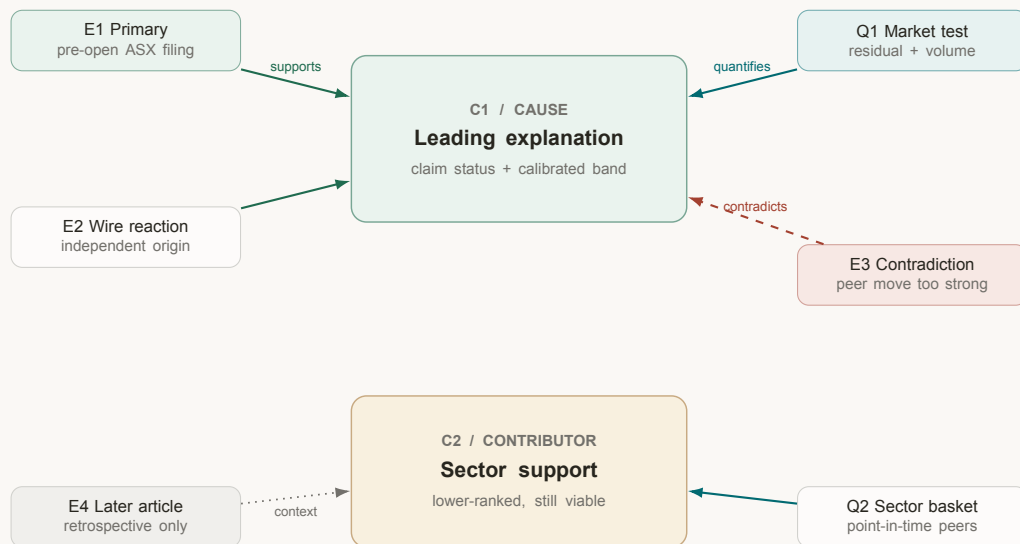


Memory boundaries derive from the project baseline [R9-R10]. LangGraph distinguishes checkpointed thread state from durable stores [R8]. NIST calls for lifecycle risk controls [R4-R5].

# Confidence is earned by observable evidence

The claim-evidence graph is the source of truth; the report is only its rendering.

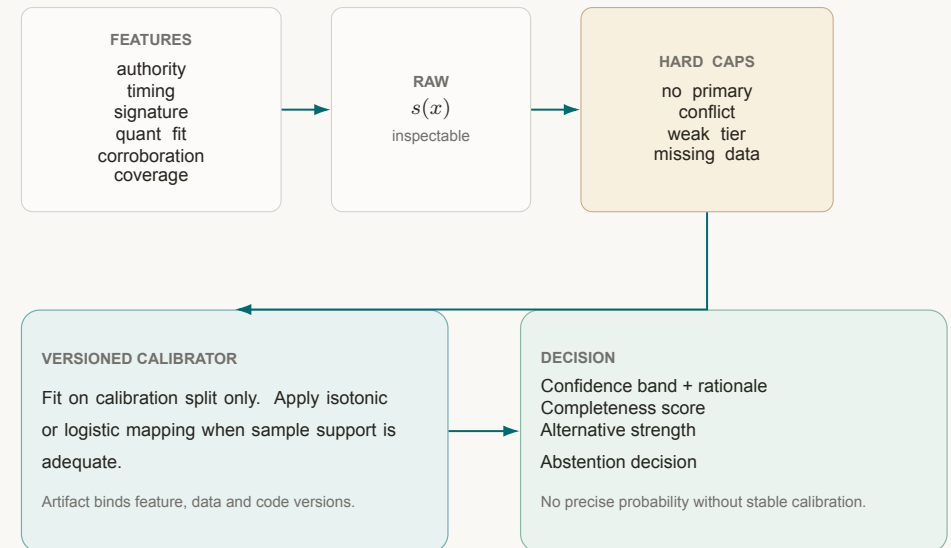
## Typed claim-evidence graph



### RELEASE RULE

A material claim with no registered support cannot ship. Retrospective evidence cannot support an earlier cause. Syndicated copies count as one origin. Conflicts remain visible.

## Confidence and abstention pipeline



### KEEP FOUR QUANTITIES SEPARATE

Anomaly score • claim support  
Selected-hypothesis confidence • investigation completeness

A dramatic move can have weak attribution. Complete retrieval can still yield no catalyst.

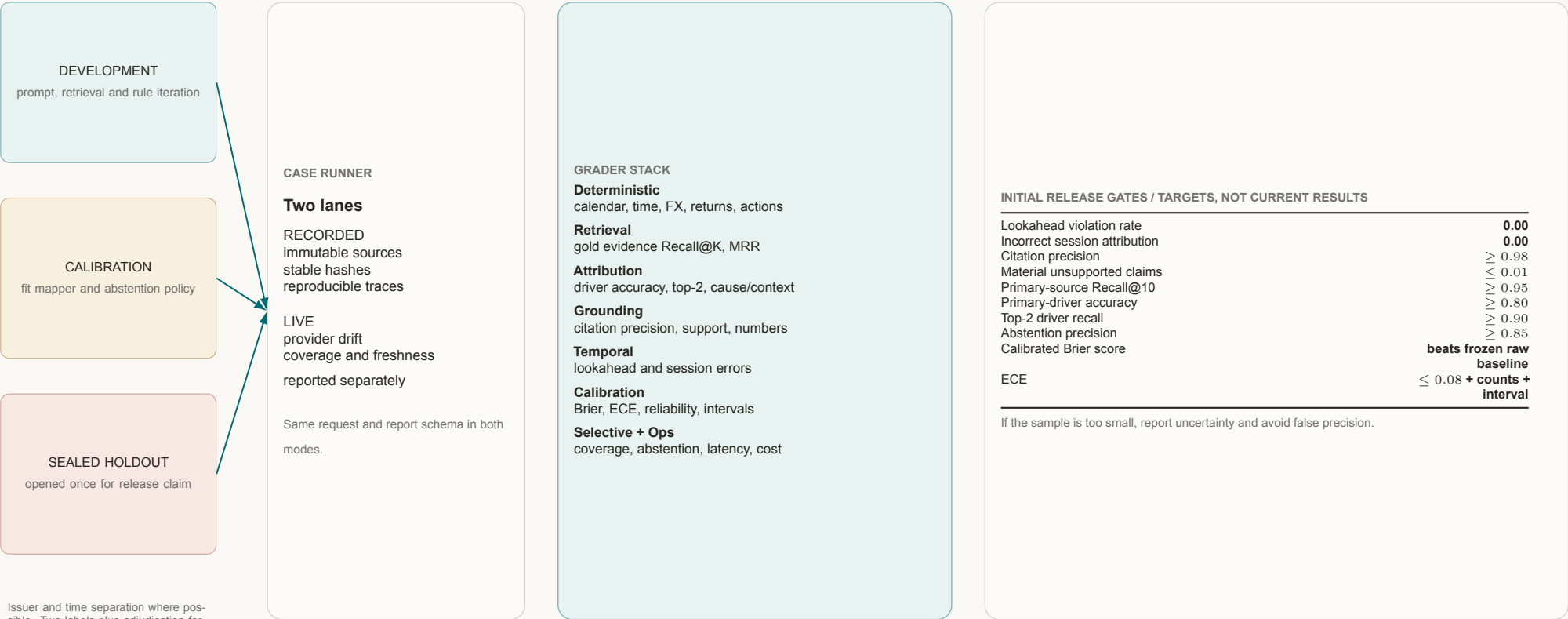
$$BS = \frac{1}{N} \sum_i (p_i - y_i)^2 \quad ECE = \sum_b \frac{|S_b|}{N} |acc(S_b) - conf(S_b)|$$

**Calibration test: predictions near 0.80 should be correct about 80% of the time in comparable cases.**

# Measure every failure surface separately

The harness must show whether a failure came from data, retrieval, reasoning, grounding, time or calibration.

## Evaluation architecture



Issuer and time separation where possible. Two labels plus adjudication for causal gold.

### FAILURE TAXONOMY

DATA • RETRIEVAL • TEMPORAL • QUANTITATIVE • REASONING • GROUNDING • CALIBRATION • INFRASTRUCTURE

Every failed case links to its trace and the first failing stage.

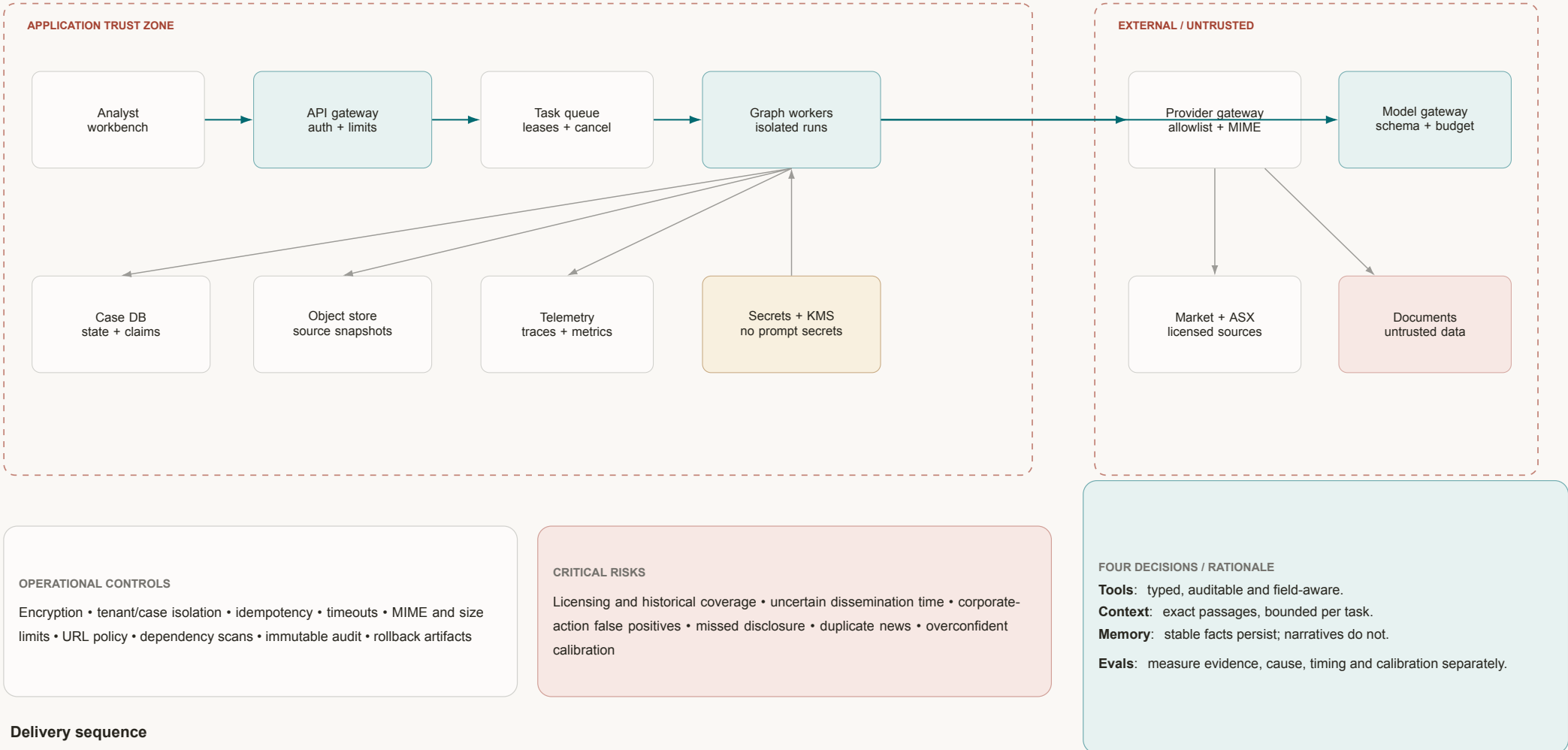
Current repository: recorded regression 1/1; no blind corpus or calibrated performance claim.



# Deployment, controls and delivery path

The production boundary protects source integrity, model access, case data and the release decision.

## Deployment and trust boundaries



## Delivery sequence



**CURRENT:** runnable MVP; 12 Python tests pass; recorded eval 1/1; confidence is explicitly **UNCALIBRATED**.

Sources: governance [R4-R5], security [R6], checkpoints [R8], roadmap [R9-R12].